L2.6b More Optimization

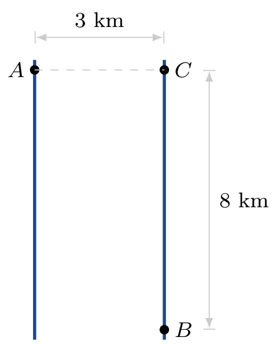
Recall: L2.6a Optimization

Name ________________________________ **Solo – 9 minutes**

1. The sum of two positive numbers is 16. What is the smallest possible value of the sum of their squares? **Do not solve it.** Write the optimization equation, the constraint, and the interval.

2. Kodi launches from A on a straight river 3 km wide and wants to reach B, which is 8 km downstream on the far bank. She rows at 6 km/h and runs at 8 km/h. Which landing point makes the trip fastest?

Do not solve it. Mark the landing point on the figure and label the two distances, then write the optimization equation, the constraint, and the interval.



3. The distance from a point to a curve involves a square root. Does the point that makes $\sqrt{x^2 + y^2}$ smallest also make $x^2 + y^2$ smallest? One sentence.